

PRADHYUMAN ARORA

Software Engineer | Cloud Infrastructure | Distributed Systems | Security
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Skills

Languages: Python, Java, TypeScript, JavaScript, C++

Cloud & Infrastructure: Azure, GCP, Docker, Kubernetes

Backend & APIs: Node.js, Express, Django, REST APIs, microservices, event-driven architecture

Databases: Azure Cosmos DB, MongoDB, Redis, MySQL

Observability & DevOps: Azure Monitor, Application Insights, Azure DevOps Pipelines

AI: LangChain, AutoGen, MCP

Experience

Microsoft — Defender for Cloud Apps

June 2024 – Present

Software Engineer, Shadow IT

- Owned backend features for the high-throughput **Cloud Discovery engine**, processing real-time logs via Azure Event Hubs to power shadow IT detection for 45k+ daily active users.
- Led the **zero-downtime migration** of 500+ GB of production data from MongoDB to Azure Cosmos DB across 12 microservices, executing cutover in under 15 minutes with 0 data loss.
- Built the Cloud App Catalog API from scratch** as a core internal shared service, exposing risk and compliance metrics for 31k+ applications to multiple engineering teams across Microsoft.
- Architected Redis-based distributed cache** (Cache-Aside pattern), slashing Azure Cosmos DB read overhead by **92%** and optimizing downstream dependency latency under peak loads.
- Engineered **AI Copilot (MCP)** for Shadow IT Discovery, integrating LLM layer with high-throughput backend APIs to enable natural language querying of SaaS security metrics.
- Designed a semantic caching layer** using **Redis Vector Search**, routing repetitive conversational queries to slash LLM inference overhead by **40%** and achieve sub-30ms latencies.
- Optimized LLM token utilization** by **85%** across the Shadow IT Copilot by implementing an MCP schema-compression proxy and response-filtering layer to strip raw JSON metadata bloat.
- Drove **zero-trust adoption** across 60+ Azure Event Hubs by migrating to Managed Identities, enforcing least-privilege access for all downstream ingestion services.
- Automated the Catalog update pipeline** via Azure DevOps, reducing lifecycle cycles from 12 to 2 hours (**83%** reduction) and saving 10+ engineering hours per release.
- Led **SFI compliance execution** across the product fleet, hardening 200+ Azure resources with Network Security Perimeters and achieving 100% closure on 12 security KPIs.
- Established **end-to-end test coverage** for the log parsing pipeline, authoring 200+ unit tests across 12 critical service paths to unblock stable weekly production deployments.
- Developed custom **MCP extensibility tools** across Azure DevOps, Outlook, and Calendar to automate internal workflows and boost team engineering velocity.

Microsoft

May 2023 – June 2023

Summer Software Engineer Intern

- Built an AI-powered phishing payload generator for Microsoft Defender for Office 365, engineering a resilient dual-GPT pipeline via Augloop for fallback reliability and automated evaluation.

AiQod

Dec 2023 – Feb 2024

Software Developer Intern

- Engineered a synthetic training data pipeline using open-source image generation models, cutting ML labeling costs by **75%** and deploying a full-stack React/Node.js object detection platform.

Education

Vellore Institute of Technology, Chennai

2020 – 2024

B.Tech. Computer Science & Engineering (Cyber Physical Systems)

CGPA: 8.96